

HAT-P-36b

R-0823

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_250223 · created 2026-07-28T15:27:30+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460729.8778 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.123 +/- 0.012
Transit depth (Rp/Rs)^2	1.51 +/- 0.29 [%]
Orbital Inclination (inc)	80.94 +/- 1.95
Airmass coefficient 1 (a1)	0.9107 +/- 0.0079
Airmass coefficient 2 (a2)	0.088 +/- 0.054
Scatter in the residuals of the lightcurve fit is	0.86 %
Best Comparison Star	#2 - [372, 209]
Optimal Aperture	3.79
Optimal Annulus	9.75
Transit Duration (day)	0.075 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.123 $\pm$ 0.012 vs 0.126 (deviation 0.25 σ)
Duration fit vs published	0.075 d vs 0.0925 d (deviation 1.46 σ)
Mid-time O-C	-15.3 min
Depth SNR	10.2
Residual scatter	0.86 %

Light curve

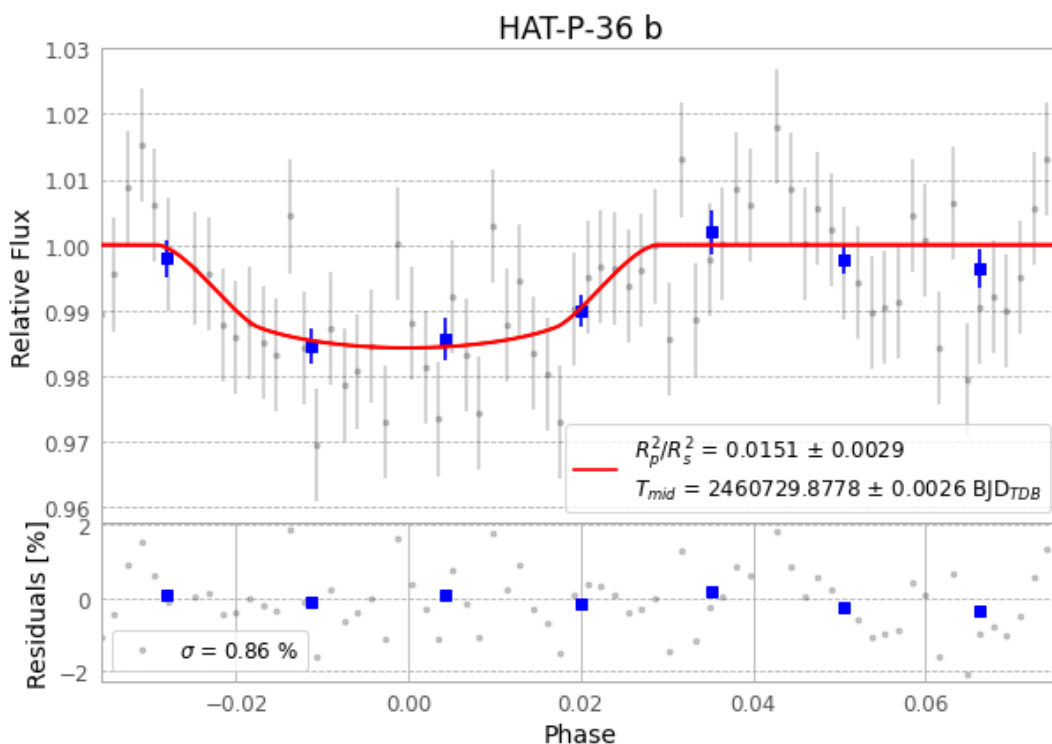


Figure 1 — FinalLightCurve_HAT-P-36 b_2025-02-23.png (SHA-256 2433a498edaabb4b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [379, 224], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4521237058b9937e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

72 FITS frames (47 MB), HATP-36250223074816.FITS → HATP-36250223112115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-250223.log (387aff2cf5fa...), FinalLightCurve_HAT-P-36 b_2025-02-23.png (2433a498edaa...), FinalLightCurve_HAT-P-36 b_2025-02-23.csv (cc3577e2a924...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 15:27Z.